

A–Ci curves

One gas-exchange measurement that reads out three different bottlenecks, and everything you have to trust to believe it.

Recorded as a question-and-answer session, 18–20 September 2026. Questions are Prabu’s, lightly tidied for spelling; answers are Claude’s, condensed from the conversation.

Give me an accessible tutorial on the A–C_i curve.

Photosynthesis can be throttled by three different things, and which one binds depends on how much CO₂ is available inside the leaf. So if you *vary* the CO₂ and watch assimilation respond, each bottleneck reveals itself in a different part of the curve. One thirty-minute measurement, three parameters.

The model behind it — Farquhar, von Caemmerer & Berry (1980), 10.1007/BF00386231 — says the leaf achieves whichever potential rate is *smallest*:

$$\begin{aligned} A_c &= V_{\text{cmax}} \frac{C_c - \Gamma^*}{C_c + K_c(1 + O/K_o)} - R_d && \text{Rubisco-limited} \\ A_j &= J \frac{C_c - \Gamma^*}{4C_c + 8\Gamma^*} - R_d && \text{RuBP-regeneration-limited} \\ A_p &\approx 3 \text{ TPU} - R_d && \text{triose-phosphate-limited} \\ A &= \min(A_c, A_j, A_p) \end{aligned}$$

That min is why the measured curve has a knee: you are watching the limitation hand off. Here Γ^* is the CO₂ compensation point in the absence of respiration, K_c and K_o are Rubisco’s kinetic constants, and O is oxygen. At saturating light $J \approx J_{\text{max}}$.

What you actually do at the bench

Clamp a healthy, fully expanded, sun-exposed leaf into a gas-exchange chamber — LI-6800, LI-6400XT, Walz GFS-3000. Set saturating light (1500–2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ for a sun leaf), fix block temperature (25 °C conventionally) and keep VPD stable. Let it acclimate at ambient CO₂ until A and g_s stop drifting — this can take 10–30 minutes and people routinely rush it.

Then step the CO₂: down first, 400 → 300 → 200 → 150 → 100 → 50; back to 400 to let the stomata recover; then up, 600 → 800 → 1000 → 1200 → 1500 (sometimes 2000). Two to three minutes per step, logging once steady.

Why down first? High CO₂ closes stomata, and they reopen slowly. If you went up first you would arrive at the low-CO₂ points with half-shut stomata — and those points are precisely the

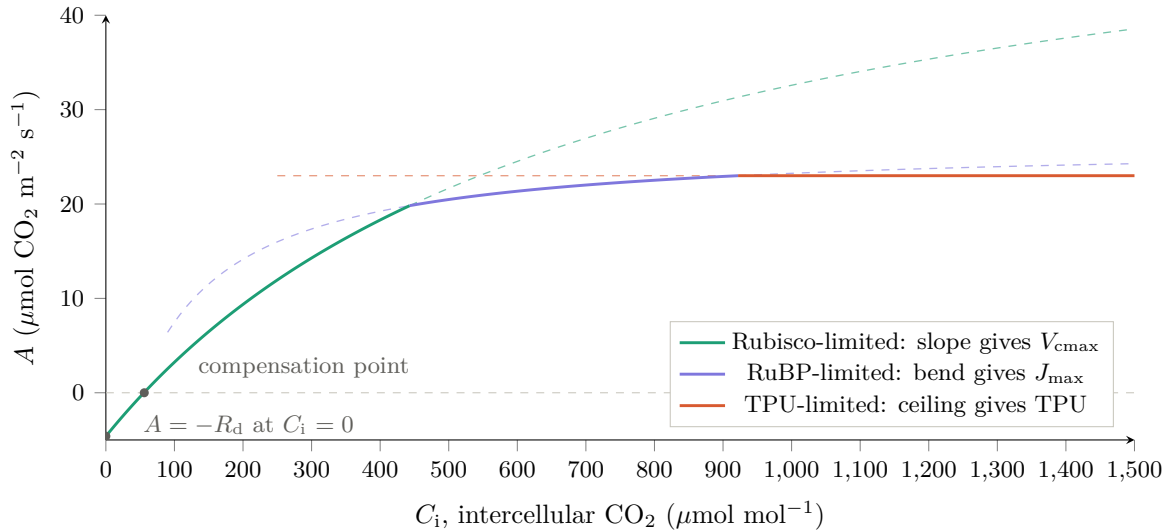


Figure 1: The measured curve (solid) is the lower envelope of three potential rates. Dashed lines continue each potential rate into the region where it is not limiting. Drawn from the Farquhar model with $V_{\text{cmax}} = 60$, $J_{\text{max}} = 110$, $\text{TPU} = 8$, $R_{\text{d}} = 1$.

ones that determine V_{cmax} . Descending keeps the stomata open while you collect the part of the curve that matters most.

Reading the curve

- **Initial rise** $\rightarrow V_{\text{cmax}}$. CO_2 is scarce, Rubisco cannot grab it fast enough, so the slope reports how much active Rubisco is present.
- **The bend flattening out** $\rightarrow J_{\text{max}}$. CO_2 is now plentiful and the limit becomes how fast electron transport regenerates RuBP.
- **The flat ceiling** $\rightarrow \text{TPU}$. The Calvin cycle outruns the leaf's ability to turn triose phosphate into sucrose and starch; phosphate gets stranded and the whole thing caps out. Sometimes A declines slightly.
- **The intercept** $\rightarrow R_{\text{d}}$. Extrapolating to $C_i = 0$, assimilation is negative, and that offset is respiration. Measure R_{dark} directly in darkness if you care about it — the extrapolated value is much less reliable.

Fitting it

Nearly everyone uses Sharkey et al. (2007), 10.1111/j.1365-3040.2007.01710.x, and its spreadsheet, or the R package `plantecophys` — Duursma (2015), 10.1371/journal.pone.0143346 — whose `fitaci()` does this in one call. Temperature normalisation to 25 °C follows Bernacchi et al. (2001), 10.1111/j.1365-3040.2001.00668.x.

The one genuinely subjective step is **assigning each point to a limitation**. Automated fitters guess the breakpoints; different guesses give different V_{cmax} . Always plot the fit with the assignments shown.

Where curves go wrong

- **Too few low- C_i points.** You want at least four below $C_i \approx 250$, or V_{cmax} rests on almost nothing.
- **Not going high enough.** Stop at 1200 and you may never reach TPU — and if TPU *was* limiting your top points, it silently corrupts J_{max} .
- **Chamber leaks.** Gaskets leak when inside and outside CO_2 differ sharply, which is exactly your endpoints. Run an empty-chamber correction.
- **Patchy stomatal closure.** Stressed leaves close in patches; the computed C_i then describes no real part of the leaf.
- **C_i versus C_c .** The equations want CO_2 at Rubisco, but you measured it in the airspace. Ignoring mesophyll conductance typically underestimates V_{cmax} by 20–30 % — fine if you are consistent, a problem when comparing against studies that corrected for it.

Sanity checks

For C_3 leaves at 25 °C: V_{cmax} roughly 20–120 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (crops at the high end, shade leaves low), J_{max} about $1.6\text{--}2 \times V_{\text{cmax}}$, R_d around 0.5–2. In Figure 1 the TPU ceiling sits at $3 \times 8 - 1 = 23$, which is where the curve goes flat. If $J_{\text{max}}:V_{\text{cmax}}$ comes out at 4, or V_{cmax} at 300, something went wrong in the fit — usually points assigned to the wrong limitation.

“So if you vary the CO_2 and watch assimilation respond” — what assimilation are you referring to?

Net CO_2 assimilation rate, the A on the y -axis: the rate at which a patch of leaf removes CO_2 from the air.

Net, because the leaf is doing two opposing things at once. Photosynthesis consumes CO_2 ; respiration and photorespiration release it. From outside you can only see the balance:

$$A = \text{gross photosynthesis} - \text{photorespiration} - \text{mitochondrial respiration}$$

Positive A means the leaf is a net carbon sink. Negative — in darkness, or below the compensation point — means net source. Gas exchange alone cannot separate the components, which is why R_d has to be measured in the dark, and why $-R_d$ appears explicitly in all three Farquhar equations: V_{cmax} , J_{max} and TPU describe *gross* capacities, and you subtract respiration to get what the instrument sees.

Is C_i the amount of CO_2 in the chamber?

No — that is C_a . C_i is the CO_2 *inside* the leaf, in the air spaces behind the stomata. There is a gradient, and CO_2 drops at every step:

C_a	chamber air — you set this: 400, 300, 200 ...
↓	stomata (g_s)
C_i	intercellular spaces — calculated; the x -axis
↓	mesophyll (g_m)
C_c	chloroplast stroma — what Rubisco actually sees

The curve is plotted against C_i rather than C_a because photosynthesis responds to the CO_2 that reaches the leaf interior, not to what you dialled into the chamber. Between the two sits stomatal conductance, which the leaf controls and which varies during the measurement.

C_i is never measured — it is inferred. Diffusion through the stomata follows Fick's law, so

$$A = g_s(C_a - C_i) \quad \implies \quad C_i = C_a - A/g_s.$$

The instrument applies a fuller version that accounts for water vapour leaving as CO_2 enters, but the principle holds: C_i is a *derived* quantity resting on measured A and g_s .

Three practical consequences:

- C_i sits well below C_a . In a healthy C_3 leaf at ambient, $C_i/C_a \approx 0.7$ — set the chamber to 400 and C_i lands near 280.
- Evenly spaced setpoints produce unevenly spaced data. You stepped $400 \rightarrow 300 \rightarrow 200$, but C_i moves by whatever the stomata allow.
- This is the weak link. Everything depends on g_s being right and uniform. With patchy stomatal closure, different parts of the leaf have genuinely different C_i , and the single computed number describes none of them.

And the last step of the gradient is the one the standard fit ignores: $C_c < C_i$, because CO_2 must still cross cell walls and membranes. Fitting against C_i when the equations want C_c is exactly the 20–30% V_{cmax} underestimate noted above.

What is g_s ?

Stomatal conductance — how readily gas passes through the stomata, the adjustable pores on the leaf surface. It is the inverse of a resistance: high g_s means pores wide open and gas moving freely. Units are $\text{mol m}^{-2} \text{ s}^{-1}$. A well-watered C_3 leaf runs about 0.1–0.6; a water-stressed leaf can drop below 0.05.

It appears in the C_i calculation because it sets how steep the CO_2 drawdown must be. For a given uptake A , narrow pores force a large gap between outside and inside — CO_2 has to be pulled far down to keep flowing through a tight opening.

It is measured through water, not CO_2 . The instrument computes g_s from transpiration: it knows the humidity of air entering and leaving, so it knows how much water vapour the leaf released, and it knows the vapour-pressure gradient driving it. Conductance is flux over gradient. The CO_2 conductance is that value divided by about 1.6. This is why leaf gas exchange measures water and carbon at once, and why A and g_s always arrive together in the data file.

g_s is under the plant's active control, responding to CO_2 , light, humidity and water status on a timescale of minutes — that is, during your measurement. Hence descending the CO_2 sequence first, and hence the fact that every C_i inherits the uncertainty in g_s . It also gives the intrinsic water-use efficiency, A/g_s : carbon gained per unit of stomatal opening.

So the LI instrument measures and reports g_s ?

It is a column in the output file, but computed, not sensed. The distinction matters because it is where errors come from.

The instrument physically measures CO_2 and H_2O concentrations entering and leaving the chamber, flow rate, leaf temperature, and chamber pressure. Everything else is arithmetic on those. Transpiration comes from the humidity difference and flow; conductance is then flux over driving gradient,

$$E = g_{\text{tw}} (w_{\text{leaf}} - w_{\text{air}}),$$

where w_{leaf} is the saturation vapour concentration at leaf temperature, assuming the internal air spaces are saturated. Invert for total conductance, remove the boundary-layer term for stomatal conductance, divide by ≈ 1.6 for the CO_2 equivalent, and C_i follows.

On an LI-6800 you see **gsw** alongside **gbw**, **gtw**, **gtc**, **A**, **E** and **C_i** ; on the older LI-6400 the column is **Cond**. A bare g_s in a paper means conductance to *water vapour* unless stated otherwise.

The chain of dependency is the point: g_s depends on leaf temperature, C_i depends on g_s , V_{cmax} depends on C_i . Three settings silently corrupt it, none of which the instrument can detect for you:

- **Leaf area**, entered by hand. If the leaf does not fill the aperture and you leave the default, every flux is scaled wrong.
- **Stomatal ratio**, which sets the boundary-layer correction. Broadleaf trees are often hypostomatous, grasses amphistomatous.
- **Boundary-layer conductance** g_{bw} , not measured per leaf but taken from a factory calibration for that chamber and fan speed.

What is transpiration?

Water evaporating out of the leaf — the plant sweating through the same pores it breathes through. Liquid water evaporates from wet cell surfaces inside the leaf, saturating the internal air spaces; outside air is drier; that gradient drives water vapour out through the stomata. In gas exchange it is E , in $\text{mmol H}_2\text{O m}^{-2} \text{s}^{-1}$ — note *millimol*, against μmol for CO_2 , because the fluxes differ by roughly a thousandfold.

It is the unavoidable cost of photosynthesis. The leaf must open its stomata to let CO_2 in, and the same opening lets water out, down a far steeper gradient. A C_3 leaf typically loses on the order

of hundreds of water molecules per CO₂ fixed, which is why drought responses show up first as g_s closing.

For the measurement it does two jobs. First, it is how g_s gets computed at all — the instrument can see conductance only because water flows through the same pores, driven by a gradient it can calculate. CO₂ cannot serve that purpose, being consumed at an unknown rate; water is merely diffusing. Second, it is the dilution correction: water vapour pouring out of the leaf dilutes the air in the chamber, slightly lowering measured CO₂. Ignore it and you would mis-attribute that dilution to photosynthetic uptake.

The driving force is the leaf-to-air vapour pressure deficit, which depends on leaf temperature — which is why thermocouple placement gets fussed over.

Is g_s different for different gases?

Yes. Same pore, different gas, different conductance: the geometry is identical, what changes is how fast each molecule diffuses through air.

Stomatal conductance: divide by 1.6. $g_{sc} = g_{sw}/1.6$. Water vapour diffuses faster than CO₂, for a reason close to Graham's law — lighter molecules move faster at the same temperature, and $\sqrt{44/18} = 1.56 \approx 1.6$. Measured diffusivities in air at 25 °C agree: about $2.5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$ for water vapour against 1.6×10^{-5} for CO₂.

Boundary layer: divide by 1.37. $g_{bc} = g_{bw}/1.37$. The still air film at the leaf surface is a second resistance in series, and transport across it is not pure molecular diffusion — it is partly stirred by airflow, so it scales with diffusivity to the two-thirds power: $1.56^{2/3} \approx 1.37$. The two then combine per gas,

$$1/g_{tc} = 1/g_{sc} + 1/g_{bc},$$

which is why the instrument reports **gsw**, **gbw**, **gtw** and **gtc** as separate columns.

This extends even to isotopes of the same gas: ¹³CO₂ is slightly heavier than ¹²CO₂ and diffuses about 4.4‰ more slowly through stomata. That discrimination is the mechanistic basis of leaf $\delta^{13}\text{C}$ as a water-use-efficiency proxy.

What is the unit of A , and how is it calculated?

$\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ — micromoles of CO₂ per square metre of leaf area per second. Micromoles because the flux is tiny; per square metre so the rate is comparable across leaves; per second because it is a rate.

The calculation is a mass balance on the air stream:

$$A = \underbrace{\frac{F(C_r - C_s)}{S}}_{\text{CO}_2 \text{ drawdown}} - \underbrace{C_s \cdot E}_{\text{dilution correction}}$$

with F the flow rate through the chamber, C_r and C_s the CO_2 entering and leaving, S the enclosed leaf area (typed in by hand), and E transpiration.

The first term is the whole idea: air goes in at 400 ppm, comes out at 380, and the 20 ppm difference times the flow rate is CO_2 per second, divided by area to make it a flux. The second term corrects for the fact that some of that drop is dilution by transpired water rather than uptake.

Worked example. Flow $500 \mu\text{mol s}^{-1}$, CO_2 falling from 400 to $380 \mu\text{mol mol}^{-1}$, leaf area 6 cm^2 :

$$(500 \times 10^{-6} \text{ mol s}^{-1}) \times (20 \times 10^{-6}) = 1 \times 10^{-8} \text{ mol CO}_2 \text{ s}^{-1} = 0.01 \mu\text{mol s}^{-1}$$

$$0.01 \mu\text{mol s}^{-1} \div 6 \times 10^{-4} \text{ m}^2 \approx 16.7 \mu\text{mol m}^{-2} \text{ s}^{-1}$$

Subtracting a dilution term of about 1.9 (see the next question) gives $A \approx 14.8$ — a healthy, well-lit C_3 leaf.

Three things that bite: leaf area is entered rather than measured, so a leaf that does not fill the aperture scales every flux wrongly with nothing looking suspicious; the drawdown must be resolvable, so flow rate has to be matched to the expected A ; and the two analysers must be matched, because you are taking a small difference between two large numbers and any offset between them lands directly in A .

What is the unit of E in that equation?

In that equation E must be in $\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}$, for the units to close:

$$C_s \cdot E = [\mu\text{mol CO}_2 \text{ mol}^{-1} \text{ air}] \times [\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}] = \mu\text{mol m}^{-2} \text{ s}^{-1} \quad \checkmark$$

which matches A . The term asks how much gas the leaf is adding to the stream, so E enters as a molar flux on the same basis as the air whose CO_2 fraction C_s describes.

But it is *reported* in $\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$ — typical values 1–10, a more human-sized number. So when taking E from a data file into this equation, divide by 1000.

The sanity check is the size of the correction. With $E = 5 \text{ mmol m}^{-2} \text{ s}^{-1}$ and $C_s = 380$:

$$\begin{aligned} \text{correct: } & 380 \times 0.005 \text{ mol m}^{-2} \text{ s}^{-1} = 1.9 \mu\text{mol m}^{-2} \text{ s}^{-1} \quad (\approx 11 \% \text{ of } A) \\ \text{wrong: } & 380 \times 5 \text{ mmol m}^{-2} \text{ s}^{-1} = 1900 \mu\text{mol m}^{-2} \text{ s}^{-1} \quad (\text{absurd}) \end{aligned}$$

A dilution correction of a few percent up to about 10 % is right. If yours comes out comparable to A itself, a factor of 1000 has slipped. Ignoring it altogether overstates photosynthesis by more than a tenth, and the error grows with transpiration — largest on hot dry days with wide-open stomata, exactly when stress responses are being measured.

Remind me of the order of measured quantities and the inferred ones.

Two groups, and the distinction matters.

Genuinely sensed: C_r and C_s , the CO_2 entering and leaving, from the two infrared gas analysers; W_r and W_s , the same for water vapour; F , the flow rate; T_{leaf} , from a thermocouple or by energy balance; chamber air temperature and pressure; and incident light.

Not sensed at all: leaf area S , the stomatal ratio, and the boundary-layer conductance g_{bw} — supplied by you, or by a factory calibration for that chamber and fan speed.

Then the arithmetic, in strict order (LI-COR's unit conventions):

1. $E = \frac{F(W_s - W_r)}{S(1000 - W_s)}$
2. $w_{\text{leaf}} = \text{saturation H}_2\text{O mole fraction at } T_{\text{leaf}} \quad \leftarrow T_{\text{leaf}} \text{ enters here}$
3. $g_{\text{tw}} = \frac{E(1000 - (w_{\text{leaf}} + W_s)/2)}{w_{\text{leaf}} - W_s}, \quad \frac{1}{g_{\text{sw}}} = \frac{1}{g_{\text{tw}}} - \frac{1}{g_{\text{bw}}}$
 $g_{\text{sc}} = g_{\text{sw}}/1.6, \quad g_{\text{bc}} = g_{\text{bw}}/1.37, \quad 1/g_{\text{tc}} = 1/g_{\text{sc}} + 1/g_{\text{bc}}$
4. $A = \frac{F(C_r - C_s)}{S} - C_s E$
5. $C_i = \frac{(g_{\text{tc}} - E/2)C_a - A}{g_{\text{tc}} + E/2}$
6. $V_{\text{cmax}}, J_{\text{max}}, \text{TPU}, R_d \leftarrow \text{fitted to all the } (A, C_i) \text{ pairs}$

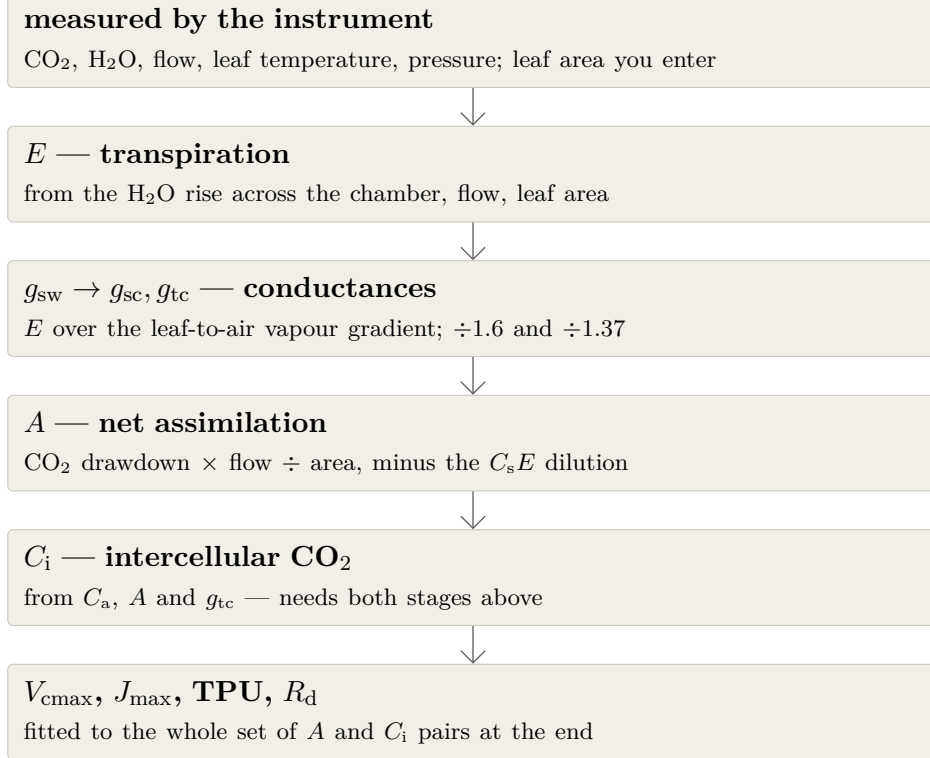


Figure 2: What the sensors see, and everything downstream that is arithmetic. An error at any stage propagates to every stage below it.

Note what depends on what. A needs only the CO_2 difference, the flow, the area and E — it never touches T_{leaf} . C_i needs A *and* the whole conductance chain, so it inherits everything. Leaf temperature enters at step 2, through the saturation vapour concentration, so the real propagation path is

$$T_{\text{leaf error}} \rightarrow \text{VPD}_{\text{leaf}} \rightarrow g_{\text{tw}}, g_{\text{sw}} \rightarrow g_{\text{tc}} \rightarrow C_i \rightarrow V_{\text{cmax}}.$$

A and E survive a bad thermocouple; C_i and everything fitted from it do not. A leaf-temperature problem shows up as a distorted x -axis, not a distorted y -axis, and that is much harder to spot by eye.

What is an IRGA?

An infrared gas analyser — the sensor that measures CO_2 and water vapour concentration. It is the heart of the instrument; everything else is plumbing around it.

CO_2 and H_2O are asymmetric molecules, so their bonds absorb infrared light at characteristic wavelengths — CO_2 strongly near $4.26 \mu\text{m}$, water vapour near $2.6 \mu\text{m}$. Shine infrared through a tube of gas, measure how much survives, and the absorption gives the concentration. Beer–Lambert, essentially, with a single-wavelength detector rather than a spectrum.

There are two of them because the whole measurement is a *difference*: a reference analyser reads the air before the leaf, a sample analyser reads it after. That difference might be 20 ppm out of 400, so any mismatch between the two lands straight in A . Hence **matching**: before a curve, the same air is diverted to both analysers and their disagreement recorded and corrected. An LI-6800 does this automatically at intervals. Skipping it is a classic way to produce a beautiful-looking curve that is quietly wrong, and on an A – C_i curve the offset is not constant across the CO_2 range, so re-matching matters.

Leaf systems are *open* — air flows through continuously and you read the steady-state difference. Closed systems seal a volume and watch concentration fall over time. Open is better here because the leaf sits at a constant, controlled CO_2 instead of a drifting one, which is exactly what you need when CO_2 is the experimental variable.

What is the difference between respiration and photorespiration?

Both release CO_2 , and that is where the resemblance ends.

Respiration is the standard cellular process — the same one running in your own cells. Sugars are broken down and oxidised, ultimately to CO_2 and water, and the energy released is captured as ATP:



Glycolysis in the cytosol, then the TCA cycle and oxidative phosphorylation in the mitochondrion. Its purpose is to power the cell — protein synthesis, transport, growth, repair. Every living cell

does it, in every tissue, day and night. It is how a plant stays alive in darkness: the plant spending the carbon it earned.

Photorespiration is a wasteful side reaction, and a consequence of one enzyme’s sloppiness. Rubisco cannot reliably distinguish CO_2 from O_2 — its full name is ribulose-1,5-bisphosphate carboxylase/*oxygenase*, and that second half is the problem. Roughly a quarter of the time in an ordinary C_3 leaf, it attaches O_2 instead of CO_2 . The product, 2-phosphoglycolate, is a two-carbon compound the plant cannot use and which inhibits other enzymes if it accumulates. A salvage pathway recovers what it can, running across chloroplast, peroxisome and mitochondrion, and in the process releases CO_2 and burns ATP and reducing power. About three-quarters of the carbon is recovered; the rest is lost. Nothing is gained — it is damage control.

	Respiration	Photorespiration
Purpose	produces usable energy	recovers from an enzymatic error
Net value	essential	a loss, $\sim 25\%$ of potential photosynthesis in C_3 at moderate temperature
When	continuously, day and night	only in light, only while photosynthesis runs
Where	mitochondrion (plus cytosol)	chloroplast + peroxisome + mitochondrion
Consumes	sugars, O_2	RuBP, O_2 , ATP
Produces	CO_2 , ATP	CO_2 , and no usable energy
Occurs in	all organisms	only photosynthetic organisms
Suppressed by high CO_2	no	yes

Why photorespiration exists at all. Rubisco evolved roughly three billion years ago, in an atmosphere rich in CO_2 and nearly devoid of O_2 . Confusing the two molecules cost nothing then. Photosynthesis itself later filled the air with oxygen, so the enzyme’s flaw is a consequence of its own success, and it has been stuck with it ever since. Evolution has improved Rubisco’s selectivity only marginally, apparently because greater specificity comes at the price of a slower reaction. Some plants route around it rather than fixing the enzyme: C_4 plants (maize, sugarcane) and CAM plants (cacti, pineapple) concentrate CO_2 around Rubisco so that oxygenation rarely happens — an advantage that grows in hot, dry, bright conditions, which is where those lineages dominate.

Where each appears in the equations above. Respiration is the explicit $-R_d$ term. Photorespiration has no term of its own — it is hidden inside Γ^* , the CO_2 concentration at which carboxylation exactly balances the CO_2 that photorespiration gives back, about $42.75 \mu\text{mol mol}^{-1}$ at 25°C . The factor $(C_c - \Gamma^*)$ says only carbon fixed above that break-even point counts, and the $4C_c + 8\Gamma^*$ denominator in A_j encodes photorespiration’s extra electron cost. So Γ^* is the *photorespiratory* compensation point, while the point where the measured curve crosses zero sits higher because it must also cover R_d : in Figure 1, $\Gamma^* \approx 43$ and the plotted compensation point ≈ 56 , the gap between them being respiration.

Sources checked against Crossref: Farquhar, von Caemmerer & Berry (1980); Sharkey et al. (2007); Bernacchi et al. (2001); Duursma (2015). Instrument details refer to LI-COR LI-6400/LI-6800 conventions.